

How Much History Should Price Private Equity?

Historical relevance in cash-flow discounting

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Abstract

How much historical data should be used to price a newly raised private-equity fund? Older vintages increase precision but may reflect different financing, valuation, and exit conditions, while recent vintages remain partly unrealized. We estimate the same cash-flow discounting model using expanding, time-decayed, and state-weighted historical samples. In a vintage-at-formation backtest, each estimator uses only information available when the target vintage is raised and is evaluated once, after that vintage matures, on its cash-flow NPV error. We also test whether NAV-based moments from still-active funds add to, or can replace, realized-cash-flow information. All choices are judged on the same later cash-flow target. **[TBD: State the headline result in NPV cents per dollar of commitment.]** The analysis treats historical relevance, rather than the number of resampled funds, as the scarce empirical object.

1 Introduction

How much historical private-capital data should be used to estimate an SDF for a fund raised today? If the pricing relation is stable, older vintages add information and full-history estimates become more precise. If it changes, the same estimator may become precisely estimated for the wrong historical mixture. More funds within a vintage can measure that vintage more accurately, but they cannot create additional economic histories.

Private equity makes this bias–variance problem unusually sharp. A long database contains only a few dozen vintage years; adjacent vintages share much of the same calendar history; and the vintages closest to the current state have not yet realized. Reaching backward increases precision but strengthens the assumption that old pricing relations remain relevant. Requiring cash-flow realization moves the usable sample away from the present. The estimation choice therefore runs along two dimensions: which history to use, and which measurements — realized cash flows are reliable but old, NAV marks are current but appraised. Resampling funds resolves neither problem.

The private-equity pricing literature has developed several ways to recover risk and performance from irregular, non-traded cash flows. Driessen et al. [2012] estimate factor exposures and abnormal performance directly from fund cash flows — the discounting framework our estimator adopts throughout — while Ang et al. [2018] recover a time series of private-equity returns from limited-partner cash flows. Korteweg and Nagel [2016] use a generalized public-market-equivalent approach to risk-adjust venture-capital cash flows. More recent work refines cash-flow benchmarking and SDF-based performance measurement [Gredil et al., 2023, Korteweg and Nagel, 2025]. These approaches differ in their pricing restrictions and treatment of illiquidity, but they largely take the historical estimation sample as given. We instead treat the use of history as the empirical choice. We call a pricing rule *historically transportable* when a rule estimated on one set of vintages also prices vintages raised at other times. Transport failure may reflect smooth drift, breaks, changes in economic-state support, or changes in fund composition. With so few vintages, the paper does not attempt an omnibus test that distinguishes all of these mechanisms. It asks which simple use of history performs best on later cash flows, following the forecasting-under- instability logic of local predictive evaluation [Pesaran and Timmermann, 2007, Rossi, 2021].

The headline design is a vintage-at-formation backtest. For each scoreable target vintage, we estimate one fixed cash-flow SDF using only information available immediately before that vintage is raised. We compare

an expanding history, exponential time decay, and time decay augmented by similarity in credit spreads, interest rates, and public-market valuations. When the target vintage later matures, each design is scored once on its realized cash-flow NPV error. This produces one genuine forecast per target vintage and directly matches the decision faced for a newly raised fund.

The same backtest prices the second dimension. At each formation date, earlier vintages that are still active contain recent but appraisal-based NAV returns. A hybrid estimator adds those short-horizon Euler moments to the cash-flow moments; a NAV-only estimator, the recency endpoint, discards realized cash flows entirely and estimates from recent marks alone. Together with the cash-flow-only rules this yields seven headline designs, bracketed by two bookends: expanding cash-flow history — maximal history, zero recency — and rolling NAV marks — zero history, maximal recency. Every design must prove itself on the target vintage’s later cash flows; NAV information is never judged by its ability to fit NAVs.

[TBD: Insert one paragraph with the main result, its magnitude, the number of scoreable target vintages, and whether the selected rule is stable to deleting one target vintage at a time.]

The paper makes two contributions. First, it provides direct evidence on how much private-equity history travels forward, separating the precision gain from a longer sample from the relevance gain of a local one. Second, it gives an appraisal-free test of whether information from still-active funds improves — or can replace — cash-flow-based pricing. The design remains informative if expanding history wins: that result would show that the variance cost of localization dominates any detectable historical drift in this sample.

2 Pricing Setup

2.1 Pricing Moments

The headline analysis fixes one SDF throughout so that differences in loss identify differences in the use of history rather than differences in factor specification. Following Driessen et al. [2012], the one-period discount factor is the inverse gross return of a levered public benchmark,

$$M_{t-1,t}(\beta) = \frac{1}{1 + r_{f,t} + \beta (R_{m,t} - r_{f,t})},$$

with $\beta \in \Theta$. Setting $\beta = 0$ prices the risk-free asset and $\beta = 1$ a self-financing public-market investment exactly, period by period: benchmark pricing is built into the functional form rather than imposed through auxiliary moments, and no separate price-of-risk parameter exists. The abnormal-return intercept of the original method is fixed at zero; Section 2.2.1 states the resulting estimand. Multifactor SDFs are robustness exercises, not additional headline designs.¹ Let $\Psi_{0,t}(\beta) = \prod_{s=1}^t M_{s-1,s}(\beta)$ denote the realized discount factor from fund inception to quarter t , with $\Psi_{0,0} = 1$, so that the time-zero capital call enters undiscounted by construction.

2.1.1 Seasoned vintages

For seasoned fund i of vintage v with committed capital K_i , net dated cash flows $CF_{t,i}$ (calls negative, distributions positive), and last observed cash flow at T_i , the realized pricing error per committed dollar is

$$g_i^S(\beta) = \frac{1}{K_i} \sum_{t=0}^{T_i} \Psi_{0,t}(\beta) CF_{t,i},$$

¹ Θ bounds β so that the levered benchmark return exceeds -100% in every sample quarter, keeping all discount factors positive.

and the seasoned vintage moment is the equal-weighted fund mean $\epsilon_v^S(\beta) = N_v^{-1} \sum_{i \in v} g_i^S(\beta)$, where N_v is the number of admissible funds in vintage v ; commitment-weighted vintage means are a pre-specified robustness variant.

The same object defines the paper’s common evaluation currency. The realized evaluation loss of a parameter β on a seasoned vintage v is

$$\widehat{U}_v^{\text{real}}(\beta) = \epsilon_v^S(\beta)^2,$$

the squared vintage-mean realized pricing error. Squaring the vintage mean, rather than averaging squared fund-level errors, targets systematic mispricing and prevents idiosyncratic fund noise — which no SDF can price — from dominating design comparisons; absolute rather than squared vintage errors are a robustness check. Because $\Psi_{0,t}$ compounds many quarterly SDF realizations, a small number of extreme discount-factor paths can dominate both estimation and evaluation. We therefore report the concentration of loss contributions across vintages and a winsorized-loss robustness variant.

2.1.2 Unseasoned vintages

For an unseasoned fund, the full NPV moment is unavailable without valuing the residual claim. The baseline bridge uses the short-horizon NAV Euler error

$$e_{t,i}^H(\beta) = M_{t-1,t}(\beta)R_{t,i}^H - 1,$$

where $R_{t,i}^H$ is the quarterly Horizon-IRR return accounting for interim calls and distributions. The unseasoned vintage moment scales the pooled fund-quarter mean into the units of the seasoned block:

$$\epsilon_v^U(\beta) = \bar{D} \cdot \frac{1}{n_v^U} \sum_{(i,t) \in v} e_{t,i}^H(\beta),$$

where n_v^U is the number of admissible fund-quarters of vintage v and $\bar{D}$ is the median discounted duration, in quarters, of seasoned funds in the estimation sample. Duration scaling is only a first-order unit conversion; it does not make the empirical scales of the two moment blocks comparable. The objective therefore applies a relative scale weight ρ_T to the unseasoned block. At each origin, ρ_T is calibrated using only older vintages that are already seasoned: their early NAV histories are treated as pseudo-young observations and compared with their full-life NPV moments. The calibration rule is fixed before forward loss is inspected; $\rho_T = 1$ and a grid around the calibrated value are robustness checks. Euler errors are never discounted back to inception and never enter the comparison score. Reported NAVs are smoothed appraisals, so this block is a candidate information bridge rather than a traded-return restriction that must hold exactly [Brown et al., 2023]. Smoothing attenuates the covariance between NAV returns and the benchmark, so a marks-based β is identified largely by the average NAV excess return relative to the realized premium in the estimation window. This is a property the backtest evaluates, not one we correct by assumption; unsmoothed returns are a robustness variant.

The main information-set comparison is deliberately parsimonious:

1. *Exclude*: estimate only with seasoned realized-cash-flow moments;
2. *Hybrid*: add short-horizon NAV Euler moments for unseasoned vintages;
3. *NAV-only*: estimate from the Euler moments alone, discarding realized cash flows — the recency endpoint, and the design closest to estimating β from recent NAV returns in practice.

All three are evaluated on the identical subsequently realized cash flows. A more complicated measurement model earns a place in the main design only if these simple comparisons demonstrate economically meaningful value from recent NAV information.

2.2 Historical Weights

At target date T , let z_v collect the economic state associated with vintage v . The baseline state vector contains three pre-performance variables: a credit spread, the interest-rate level, and a public-market valuation ratio. The variables are measured at or immediately before vintage formation and standardized using the estimation sample only. State conditioning also serves as a reduced-form hedge against time-varying premia on factors the benchmark omits: it localizes estimation in states where such premia are plausibly similar (Appendix C).

For historical vintage v , define the unnormalized relevance weight

$$w_{v,T}(h, b) = \exp\left[-\log(2)\frac{T-v}{h}\right] \exp\left[-\frac{1}{2b^2}(z_v - z_T)'(z_v - z_T)\right],$$

where h is a time half-life and b is a common state-similarity bandwidth. The common bandwidth avoids estimating three separate state sensitivities from a few dozen vintages. Setting $h = \infty$ removes time decay and setting $b = \infty$ removes state conditioning.

Let q_v be a reliability weight, set to the vintage fund count N_v in the baseline; vintage-equal weighting ($q_v = 1$) is a pre-specified robustness variant. The products $w_{v,T}q_v$ are normalized to sum to one jointly across all vintages entering the objective, seasoned and unseasoned alike, so that a vintage's relevance does not depend on the block through which it enters. For historical weighting rule W and information set $A \in \{\text{Exclude, Hybrid, NAV}\}$, estimate

$$\widehat{\beta}_T(W, A) = \arg \min_{\beta \in \Theta} \left\{ \mathbf{1}\{A \neq \text{NAV}\} \sum_{v \in \mathcal{V}_T^S} \widetilde{w}_{v,T} \epsilon_v^S(\beta)^2 + \rho_T^A \sum_{v \in \mathcal{V}_T^U} \widetilde{w}_{v,T} \epsilon_v^U(\beta)^2 \right\},$$

where ρ_T^A is 0 under Exclude, the calibrated ρ_T under Hybrid, and 1 under NAV-only, whose single-block objective makes the scale irrelevant. We collect a complete headline design in $c = (W, A)$ and write $\widehat{\beta}_T(c)$ below. Because unseasoned vintages are the most recent, time decay concentrates weight on the Euler block. Every hybrid configuration therefore reports its effective Euler share — the scale-adjusted weight share $\pi_U(T) = \rho_T \widehat{W}_U / (\widehat{W}_S + \rho_T \widehat{W}_U)$ with $\widehat{W}_B = \sum_{v \in \mathcal{V}_T^B} \widetilde{w}_{v,T}$ — which makes this interaction visible rather than leaving it implicit in the kernel and the scale weight.

The seven headline designs combine three historical weighting rules with the three information sets:

1. **Expanding:** equal weight on the full available history;
2. **Decay:** exponential time weights with a fixed baseline half-life of 8 vintage years, roughly half a typical fund life;
3. **Time-state:** the same decay combined with similarity in credit spreads, interest rates, and public valuations.

Exclude and Hybrid are each crossed with all three weighting rules; NAV-only enters once, under the decay weights, so that its comparison with the corresponding cash-flow rules isolates the information set rather than the window shape.

Rolling windows, alternative half-lives and bandwidths, NAV-only under time–state weights or a hard trailing window, multifactor SDFs, and a regularized time-varying parameter path are robustness exercises [Cui et al., 2024]. They do not enlarge the headline selection set.

Every candidate must satisfy two feasibility restrictions before it can be scored: the weighted moment Jacobian must be sufficiently well conditioned, and the weight concentration must leave an effective vintage count

$$V_{\text{eff}}(T) = \frac{1}{\sum_v \tilde{w}_{v,T}^2}$$

comfortably above the number of SDF parameters. The count is computed both over all vintages and, with weights renormalized within the block, over the seasoned block alone; whenever a design uses cash-flow moments, the threshold must hold for the seasoned block, since realized cash flows anchor identification. NAV-only designs face an analogous screen: a minimum effective number of fund-quarters and a minimum realized variation of the benchmark excess return within the weighted window, because a window whose realized premium is close to zero identifies β only through noise in smoothed marks (Appendix B). These screens prevent an apparently local estimator from being driven by one historically similar vintage, propped up by appraisal-based moments alone, or fitted to a window without price variation.

2.2.1 Pricing law versus performance measurement

The fitted β is a predictive discount-rule parameter, not a pure risk exposure or an estimate of abnormal performance. With the intercept fixed at zero, average in-window performance and unspanned premia can be absorbed into the fitted exposure. We therefore interpret $\hat{\beta}$ only as the leverage on the public benchmark that best rationalizes the chosen historical sample. In-window residuals are diagnostics of fit; all performance statements in the paper concern target vintages excluded from estimation. An appendix derives the first-order premium channel and reports it as a specification diagnostic, not as a recovery of structural α and β .

2.3 Historical Transportability

2.3.1 The transportability matrix

Let s index a historical estimation window and t an excluded, fully realized evaluation vintage. Throughout this subsection s denotes the window’s information end date; where present, the time-decay and state kernels are centered on s and z_s , rather than on a target vintage, and each row uses a single information set, $A \in \{\text{Exclude, NAV}\}$. Define the NAV-free transport loss

$$L_{s \rightarrow t}(c) = \hat{U}_t^{\text{real}}(\hat{\beta}_s(c)),$$

where the evaluation score uses only realized fund cash flows. Omitting the hybrid keeps every row’s information set pure: cash-flow-estimated and marks-estimated windows can be compared directly. Stacking these losses over estimation windows and evaluation vintages yields a *historical transportability matrix*. Its rows are window SDFs, its columns are evaluation vintages, and each entry measures how well a pricing law travels across economic time; the NAV rows show how far marks-estimated rules travel into realized cash flows, the descriptive counterpart of the paper’s fourth question.

Aggregating by the signed temporal distance $d = t - s$, measured from the window end, gives the transport loss curve

$$\mathcal{T}_c(d) = \frac{1}{N_d} \sum_s L_{s \rightarrow s+d}(c).$$

Positive distances measure forward transportability, the only direction with a real-time interpretation; backward entries are diagnostics of structural stability, not feasible investment rules. The matrix is descriptive: a flat surface is consistent with broad transportability, loss increasing in $|d|$ with temporal localization, and isolated high-loss columns with historically unusual target periods. None of these patterns alone proves a particular structural-break model.

2.3.2 Calendar-time overlap

Adjacent vintage moments share public-factor histories because their funds remain active for many years. We therefore measure pairwise calendar-time overlap rather than treating vintage labels as independent by construction. Let $a_{v,t}$ denote vintage v 's absolute discounted cash-flow exposure to calendar period t , evaluated at reference parameter values fixed in advance.² Define

$$O_{vv'} = \frac{\sum_t a_{v,t} a_{v',t}}{\sqrt{\sum_t a_{v,t}^2} \sqrt{\sum_t a_{v',t}^2}}.$$

The overlap matrix is reported alongside the transportability matrix. Estimation–evaluation pairs with overlap above a pre-specified threshold are flagged, and headline comparisons are re-reported excluding them. Because complete separation would eliminate most of the data, overlap is quantified and stress-tested rather than treated as a binary admissibility standard.

2.4 The Vintage-at-Formation Backtest

Each scoreable target vintage v receives one forecast. Let τ_v be the last information date before vintage v is formed. The estimator $\hat{\beta}_v(c)$ uses only cash flows, NAVs, public factors, and state variables dated no later than τ_v ; its time and state kernels are centered on v and z_v . The target vintage never enters its own estimator. Under the hybrid and NAV-only treatments, only earlier vintages that were still active at τ_v contribute NAV Euler moments.

After target vintage v becomes eligible for cash-flow-only evaluation, its forward loss is

$$L_v(c) = \hat{U}_v^{\text{real}}(\hat{\beta}_v(c)).$$

The rule is therefore estimated at the same point at which it would have been used and the target is scored exactly once. This eliminates repeated use of the same eventual outcome across adjacent origins and directly matches the current application: estimate a pricing rule immediately before a new vintage is raised.

2.5 What the Backtest Can Establish

The vintage-at-formation exercise is pseudo-real-time rather than literal real time. The final extract does not reconstruct historical database coverage, later backfills, or every field's original reporting date. We use timestamps where available and exclude suspiciously late first reports, but the remaining limitation is stated directly.

Maturity creates a second limit: the last scoreable target vintage precedes the final sample date by roughly one fund life. Applying the selected rule to today's vintage therefore remains an extrapolation beyond the last directly validated cohort. We report the last scoreable vintage and whether the current state lies inside the historical support used by the state kernel.

²A Jacobian-based exposure measure yields an alternative overlap matrix and is reported as a variant.

2.6 Forward Selection Criterion

Let $\mathcal{V}^{\text{eval}}$ denote the target vintages that can be estimated at formation and subsequently evaluated on the common cash-flow-only criterion. For headline design $c = (W, A)$, define

$$\widehat{R}(c) = \frac{1}{|\mathcal{V}^{\text{eval}}|} \sum_{v \in \mathcal{V}^{\text{eval}}} L_v(c).$$

All seven designs face the same target vintages, so comparisons are paired. Fixed-design losses answer the four substantive questions: whether decay beats expanding history, whether state similarity improves on decay, whether the NAV bridge improves on the corresponding cash-flow-only rule, and whether NAV moments alone — the recency endpoint — can replace realized cash flows entirely.

Selecting the lowest loss from the same backtest creates a winner’s curse. The preferred current rule is therefore the simplest design within one standard error of the minimum and within the model confidence set [Hansen et al., 2011]. We also evaluate the selection rule sequentially. At formation of target vintage v , selection may use only earlier target vintages whose evaluation outcomes would already have been observable at τ_v . The resulting sequence measures what the complete rule, including selection, would have delivered historically. Fixed-design comparisons and sequential selection are reported together; neither is described as a clean estimate of the other’s performance.

2.7 Inference and Reporting

Model comparisons use target-vintage paired loss differences [Giacomini and White, 2006]. We report root-mean-squared NPV error in cents per dollar of commitment, the percentage change relative to expanding history, and the signed errors behind the loss. Inference uses calendar-time blocks that respect overlapping fund lives, supplemented by leave-one-target-vintage-out stability. The number of target vintages and funds is shown beside every comparison. With only a few economic histories, uncertainty intervals and the model confidence set measure precision; they are not treated as a binary test of structural stability.

3 Data and Empirical Design

This section defines the cash-flow-only target, the information available at each target vintage’s formation, and the seven headline designs.

3.1 Private-Capital Cash-Flow Data

The fund data are drawn from [TBD: data provider and extraction date]. The unit of observation is a dated limited-partner cash flow for a [TBD: buyout / broader private-capital] fund. Required fields are fund identifier, vintage year, commitment, dated capital calls, dated distributions, reported NAVs, strategy, geography, and — where available — general-partner identifier. Cash flows and NAVs are measured net of management fees and carried interest from the limited partner’s perspective and converted to [TBD: currency and real/nominal units]. We exclude funds with internally inconsistent cash-flow histories, missing commitments, nonpositive commitments, or implausible scale ratios under a pre-specified cleaning rule. The analysis retains all admissible funds within a vintage; it does not create pseudo-vintages by resampling funds.

The first data table should make the thin time-series dimension visible rather than reporting only a large fund count:

Table 1: Private-capital sample by vintage period

Vintage period	Vintages	Funds	Commitments	Funds/vintage	Liquidated share
[TBD: early period]					
[TBD: middle period]					
[TBD: recent period]					
Full sample					

Notes: Commitments are reported in [TBD: units]. Funds per vintage is the median across vintage years. Liquidated share is the fraction of funds with zero residual NAV at the final sample date.

3.2 The Cash-Flow-Only Evaluation Sample

Estimation and evaluation use different strictness. The evaluation target uses only funds recorded as liquidated with zero residual NAV: their loss contains calls and distributions only, and neither reported NAV nor an estimated terminal value enters the score. The seasoned cash-flow moments used in estimation instead admit *near-realized* funds, those whose residual NAV per dollar of commitment satisfies

$$r_{i,\tau} = \frac{\text{NAV}_{i,\tau}}{K_i} \leq \delta_K,$$

with a pre-specified small threshold [TBD: **proposed baseline: 5%**]. Unlike the residual share of total value, this rule directly bounds the omitted appraisal per dollar of commitment, and it moves the freshest usable estimation vintage several years closer to each formation date. The bounded truncation reaches every design through the same funds and largely cancels in paired comparisons. Liquidated-only estimation and a near-realized evaluation expansion are robustness variants, each labelled as such and never described as fully realized.

A fund excluded from evaluation may still contribute to the Euler block of the hybrid and NAV-only designs at an earlier origin. For every target vintage we report the available seasoned and unseasoned funds, the number of liquidated evaluation funds, and the share of vintage commitment represented by the cash-flow-only target. This makes early-liquidator selection and coverage differences visible. The descriptive feasibility audit reports the number of scoreable target vintages under each sample rule before the design grid is frozen.

3.3 Public Factors and the Baseline SDF

Public-market returns and the risk-free rate are obtained from [TBD: **source**] at the same frequency as the cash-flow discounting exercise. The headline design uses the freely estimated one-factor β of Section 2.1 for all seven candidates. Risk-free discounting, the $\beta = 1$ public-market-equivalent benchmark, and a low-dimensional multifactor model are reported as robustness checks. Factor definitions, currency, return convention, and the mapping from quarterly factors to dated cash flows are fixed before forward pricing losses are inspected.

3.4 Economic-State Variables

The state vector z_v contains one variable for each of three economic dimensions: a corporate credit spread, the interest-rate level, and a public- equity valuation ratio. The proposed baseline definitions are the high-yield option-adjusted spread, a long-term nominal government yield, and the cyclically adjusted price–earnings ratio [TBD: or a price–dividend ratio]. To avoid using fund outcomes to define similarity, each variable is measured as its average over the 12 months ending immediately before the vintage year. At each origin, levels are standardized using only the information set at formation. We report whether each target state z_v , and the final current state, lies inside the available historical support. A target outside that support is an extrapolation problem and should not receive a falsely precise local estimate.

3.5 Target-Vintage Information Sets

Target vintages begin with the first formation date at which all seven headline designs satisfy the effective-vintage, Jacobian, and NAV-window conditions. For target vintage v , every input is truncated at τ_v , immediately before formation. The target’s pre-formation state z_v is observable, but none of its fund outcomes are. The same target vintages and evaluation funds are used for every paired comparison. Sequential selection may use only evaluation outcomes that would already have become observable by τ_v .

3.6 Pre-Specified Design Grid

The grid is intentionally coarse. Its final endpoints are fixed after a descriptive feasibility audit, but before comparing out-of-sample losses.

Table 2: Headline empirical design

Object	Baseline candidates	Restriction or robustness
SDF	Estimated one-factor β	Fixed across all seven headline designs
Historical weighting	Expanding; exponential decay; time–state	One incremental comparison at each step
Time half-life	8 vintage years	4 and 12 years in robustness
State bandwidth	1.5 standardized units	0.75 and 3 in robustness
Information set	Exclude; NAV Euler hybrid; NAV-only (decay)	Common later realized-cash-flow target
NAV-block scaling	Prior-origin pseudo-young normalization	$\rho = 1$ and neighboring values in robustness
Estimation moments	Near-realized funds ($r_{i,\tau} \leq \delta_K$)	Liquidated-only estimation in robustness
Evaluation sample	Liquidated funds with zero residual NAV	Near-realized expansion reported separately
Feasibility	$V_{\text{eff}} \geq 8$ incl. seasoned block; NAV screen	Jacobian conditioning reported separately

4 Empirical Results

4.1 The Effective Historical Sample

Table 1 first documents the nominal and effective sample size. A vintage-by-calendar-year cash-flow heat map shows the long overlap of fund lives, while the overlap matrix O summarizes common factor exposure. We report the eigenvalue-based effective temporal rank and effective vintage counts under each candidate weighting rule. This section establishes how much independent economic history is actually available before asking whether old history remains relevant.

4.2 Ex-Post Historical Transportability

The first main figure is the historical transportability matrix for the baseline SDF. Rows identify estimation windows, columns identify excluded realized vintages, and colors represent realized pricing loss; cash-flow-estimated and NAV-estimated rows appear in the same figure. Signed-distance loss curves then summarize forward and backward transport. The empirical questions are whether loss rises with temporal distance, whether the pattern is asymmetric, and whether particular evaluation vintages are difficult for every historical window. This figure is descriptive; the vintage-at-formation backtest supplies the directional evidence.

4.3 How Much History Travels Forward?

The first main comparison holds the one-factor SDF and information set fixed and compares expanding history with exponential decay on the same target vintages. It reports root-mean-squared NPV error in cents per dollar of commitment, the paired difference, V_{eff} , and leave-one-target-vintage-out stability. This is the paper’s primary answer to whether local history improves pricing of later cash flows.

4.4 Does Economic-State Similarity Add Information?

We next compare time decay with the time–state rule. Weight plots show which distant vintages are recovered because their credit spreads, interest rates, or public valuations resemble the target. Leave-one-state-variable-out results are secondary diagnostics. State conditioning is preferred only if it improves forward loss without collapsing the effective vintage count.

4.5 Do NAV Marks Add to or Replace Cash-Flow Information?

For every target vintage, we compare exclusion, hybrid, and NAV-only estimators under the same decay weights and on identical later cash flows. The table reports paired loss differences, the calibrated ρ_v , the effective Euler share, and sensitivity to subsequent NAV revisions. The hybrid column answers whether marks add information at the margin; the NAV-only column answers whether the most current information set can stand alone, and how often it fails the benchmark-variation screen. This distinguishes useful early information from a mechanical fit to smoothed reported values.

4.6 The Preferred SDF and the Pricing of Recent Vintages

The final results section applies the simplest design within one standard error of the minimum and the model confidence set at the current information date. We report its benchmark exposure, every historical vintage weight, the last directly scoreable target vintage, and whether today’s state lies inside historical support. The expanding-history estimate is shown alongside it. This presentation distinguishes what is validated historically from the final extrapolation to the current vintage.

5 Robustness and Method Validation

5.1 Alternative Sample and Moment Choices

The near-realized expansion varies the residual-NAV-per-commitment threshold; the strict liquidated-fund result remains the headline. We also remove crisis and pandemic vintages one at a time, alter fund-size reliability weights, and restrict the sample by strategy or geography where coverage permits. Vintage-equal weighting checks that data-rich vintages do not determine the conclusion.

5.2 Alternative Windows, States, and SDFs

The appendix restores rolling windows and the full half-life and bandwidth grid, varies the NAV precision weight, re-estimates NAV-only under time–state weights, a hard trailing five-year window, and unsmoothed returns, and compares risk-free, PME, multifactor, and regularized time-varying SDFs. We also vary the interest-rate and valuation proxies. These exercises test whether the four headline incremental comparisons survive reasonable alternatives without enlarging the main selection set.

5.3 Placebos and Method Validation

Two placebos discipline the interpretation of state similarity. First, state vectors are permuted across vintages while their time ordering is retained. Second, the target state is replaced by a future state that was unavailable at the origin. Genuine transport gains should disappear under the first placebo; the second quantifies how much an inadvertently forward-looking state construction could exaggerate performance. A compact appendix calibration matches the empirical vintage count, cash-flow-age profile, and calendar overlap under a stable and a smoothly drifting SDF, with and without appraisal smoothing. Its limited purpose is to verify that the procedure does not mechanically prefer localization or the NAV bridge when the law is stable.

6 Conclusion

Private-capital SDF estimation cannot solve a thin and delayed history by resampling funds more aggressively. The economically relevant decision is how much old history to use for a newly raised vintage and whether current but appraised NAV information should supplement — or replace — realized cash flows. The vintage-at-formation backtest answers these questions on later cash flows, with one prediction per target vintage, one fixed baseline SDF, and two bookends: expanding cash-flow history and rolling NAV marks.

[TBD: Summarize the four incremental findings: decay versus expanding history, state similarity versus decay, hybrid versus exclusion, and NAV-only versus the cash-flow-anchored rules. State the selected rule and its uncertainty.] If localization wins, the evidence shows that the relevance gain from recent or economically similar vintages exceeds its precision cost. If expanding history remains competitive, the conclusion is equally useful: the available sample is too thin for more adaptive estimators to improve reliably. In either case, the current estimate is presented with the historical vintages that support it and the distance from the last cohort on which the rule could be directly validated.

A NAV-Block Scale Calibration

The baseline duration scaling puts quarterly Euler errors approximately in lifetime-NPV units, while ρ_T normalizes their empirical dispersion. At origin T , let $\mathcal{P}_T$ contain only vintages that are already liquidated. For each $u \in \mathcal{P}_T$, reconstruct the NAV Euler moment that would have been observed after a fixed early-life horizon, and denote its duration-scaled value at the reference $\beta = 1$ by x_u . Let y_u be the corresponding full-life cash-flow NPV moment at the same reference parameter. Using robust cross-vintage variance estimates computed only within $\mathcal{P}_T$, set

$$\rho_T = \frac{\widehat{\text{Var}}_T(y_u)}{\widehat{\text{Var}}_T(x_u)}.$$

The numerator and denominator are floored away from zero, and the allowable range of ρ_T is fixed before forward loss is inspected. This calibration equalizes the historical dispersion of the two blocks; it is not an estimate of structural measurement precision, does not treat NAV returns as traded returns, and does not use the target vintage’s eventual outcome.

B Extended Design Set

The appendix sensitivity set adds rolling windows of 8, 12, and 16 vintages; time half-lives of 4 and 12 years around the 8-year baseline; state bandwidths of 0.75 and 3 around the 1.5-unit baseline; and NAV-only variants under time–state weights, a hard trailing five-year window, and unsmoothed NAV returns. It also reports risk-free discounting, the $\beta = 1$ PME benchmark, and a low-dimensional multifactor SDF. These variants are compared with the headline designs but do not enter the headline seven-model selection set.

The NAV-window screen requires, within the weighted estimation window, a minimum effective number of fund-quarters, $n_{\text{eff}} = 1/\sum_{(i,t)} \tilde{\omega}_{i,t}^2$ computed from the normalized fund-quarter weights, and a minimum weighted variance of the benchmark excess return. Both thresholds are fixed in the descriptive feasibility audit before any forward loss is inspected. Windows failing the screen are reported as infeasible rather than estimated, since a window in which the realized premium is close to zero identifies β only through noise in smoothed marks.

As a higher-dimensional check, the regularized time-varying path solves

$$\min_{\{\beta_v\}} \sum_v \epsilon_v(\beta_v)^2 + \kappa \sum_{v>1} (\beta_v - \beta_{v-1})^2.$$

At each sequential target vintage, κ may be chosen only from evaluation outcomes already observable by that vintage’s formation date. The method is reported as a check on whether simple weighting misses economically important smooth drift, not as an additional headline candidate.

C Structural Interpretation Diagnostic

Fixing the abnormal-return intercept at zero makes the fitted exposure a composite. In a first-order return representation with true abnormal component α , exposure β , and included benchmark premium λ_s in estimation window s , the zero-intercept fit satisfies

$$\beta_s^* \approx \beta + \frac{\alpha}{\lambda_s}.$$

For an evaluation window t , the corresponding signed residual is approximately

$$\alpha \left(1 - \frac{\lambda_t}{\lambda_s}\right).$$

Thus, even a stable α can generate transport loss when included-factor premia differ across windows. Omitted unspanned premia create the same observational problem. Plots of signed residuals against realized benchmark premia are therefore specification diagnostics only: they cannot by themselves separate skill, omitted factors, and changes in the pricing relation.

D Compact Method Validation

The calibration matches the empirical vintage count, cash-flow-age profile, and calendar overlap under a stable and a smoothly drifting SDF, each with and without appraisal smoothing. Its purpose is limited to checking that the selection procedure does not mechanically prefer localization or the NAV bridge when the pricing law is stable.

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